



Treehouse Design

rivt Report

Example report

Proj. 0001

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02.1 Table test

Table 1: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL

02.2 overview

This report covers the structural design of a treehouse in Novato, California. The design follows the requirements of the California Building Code (CBC).

The treehouse is supported by a single Douglas fir beam spanning 12 feet between two trees. The beam supports a uniformly distributed load (UDL) from the treehouse floor and live load from occupants. The design includes calculations for the required beam size and material properties to ensure safety and compliance with building codes.

02.3 Load Combinations

Dead and live load contributions to beam UDL.

Table 2: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

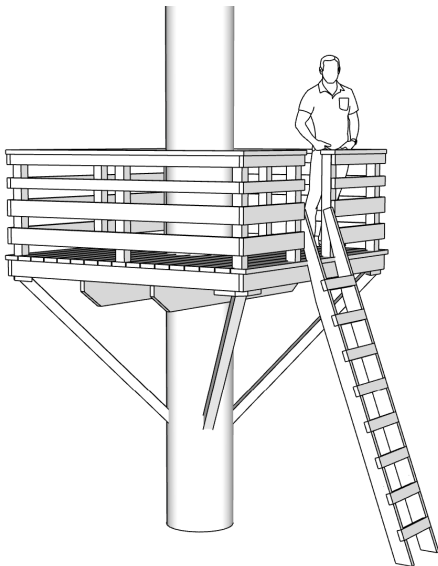


Fig. 1 - Treehouse